Elif Bilge

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Experience

Computer Vision Research Engineer, Disun Software – Izmir, TR (Remote) March 2022 - Oct 2025

Led development of medical image analysis algorithms for pathology and radiology applications, implementing deep learning models for segmentation, classification. Managed end-to-end project lifecycle from research to deployment, including TUBITAK-funded projects.

- **Mediastinal Lymph Node Segmentation & Station Classification:** Developed an algorithm to capture spatial relationships between lymph nodes and anatomical structures for IASLC-based mapping, resolving multi-station node ambiguities via specialized feature engineering—enhancing cancer staging accuracy, standardizing classification across institutions, and reducing radiologist workload.
- **Tumor Budding Segmentation:** Developed a specialized neural network pipeline for tumor budding segmentation in H&E images, addressing the subjective nature of this task for pathologists. Integrated multi-resolution ROI analysis to detect tumor budding candidates and reduce false positives in large-scale H&E images, and deployed the solution in a user-friendly interface for pathologist evaluation.
- **HE2DAPI Multi-Modal Image Registration:** Developed a keypoint-based image registration pipeline to align corresponding regions between H&E and DAPI-stained whole-slide images. Leveraged complementary information from both staining modalities to improve cell type identification and tissue morphology visualization. Enhanced diagnostic accuracy and detection of pathological abnormalities through precise spatial alignment. Developed a GUI to facilitate interactive testing, visualization, and evaluation of the registered regions by pathologists.

Research Engineer, National Magnetic Resonance Research Center – Ankara, TR Feb 2021 – Dec 2022

- **FPGA-Based MRI Spectrometer Development:** As a research engineer, conducted foundational research and authored the project proposal for a national MRI development initiative in collaboration with ASELSAN aimed at eliminating vendor dependency. Focused on MRI spectrometer design and pulse sequence development to enable seamless communication between sequences and the scanner for flexible waveform transmission.

Computer Vision Research Engineer, Papilon Defense – Ankara, TR July 2020 – Nov 2020

- **Automated License Plate Recognition System:** Designed and implemented a real-time license plate detection and OCR pipeline for smart parking systems, utilizing a YOLO-based object detector coupled with custom-trained character recognition models for robust and accurate performance.

Electronic Engineer, Intern, ASELSAN – Ankara, TR Jan 2018 – Apr 2018

- **Web Service Architecture Development:** Implemented secure user authentication system using Windows Communication Foundation (WCF) framework with MVC design pattern.

Electronic Engineer, Intern, KIWI – Ankara, TR July 2017 – Aug 2017

- **Drone Motor Control System (FPGA, VHDL):** Developed FPGA-based communication protocol for precision drone motor control systems.

Education

Bilkent University, B.S. in Electrical and Electronics Engineering Sept 2015 – Feb 2020

Technical Skills

Programming Languages: Python, MATLAB, C#, VHDL, SQL
ML/DL libraries: PyTorch, TensorFlow/Keras, Scikit-Learn, OpenCV, Scipy, NumPy, Pandas, SimpleITK, PyDicom, ITK-SNAP, 3D Slicer, MONAI
Tools & Platforms: Linux, Docker, Git, WandB, MLflow, CUDA, AWS
Languages: English (C1), German (B1), Turkish (Native)